

Embedded systems exp 30

```
#include <systemc.h>

SC_MODULE(rtos_scheduler) {

    void task1() {
        while(true) {
            cout << "Task1 Running at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void task2() {
        while(true) {
            cout << "Task2 Running at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(rtos_scheduler) {
        SC_THREAD(task1);
        SC_THREAD(task2);
    }

};

int sc_main(int argc, char* argv[]) {
    rtos_scheduler obj("Scheduler");
    sc_start(6, SC_SEC);
}
```

EDAplayground

NewRunSaveCopy

ResourcesCommunityHelpPlaygroundsProfile

Brought to you by

DOULOS

DOULOS does not endorse training material from other suppliers on EDA Playground.

Languages & Libraries

Testbench + Design

C++/SystemC

Libraries

None

SystemC 2.3.3

Tools & Simulators

C++

Compile Options

DSC_INCLUDE_FX

Select

☐ Use -pedantic -Wall -Wextra

Run Options

Run Options

☐ Use run.bash shell script

☐ Open EPWave after run

☐ Show output text file

☐ Show output SVG file

☐ Download files after run

testbench.cpp

```
10 void task2() {
11     while(true) {
12         cout << "Task2 Running at "
13         << sc_time_stamp() << endl;
14         wait(1, SC_SEC);
15     }
16 }
17 SC_CTOR(rtos_scheduler) {
18     SC_THREAD(task1);
19     SC_THREAD(task2);
20 }
21 }
22 int sc_main(int argc, char* argv[]) {
23     rtos_scheduler obj("Scheduler");
24     sc_start(6, SC_SEC);
25     return 0;
26 }
27 }
```

design.cpp

```
1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
```

LogShare

Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
Done